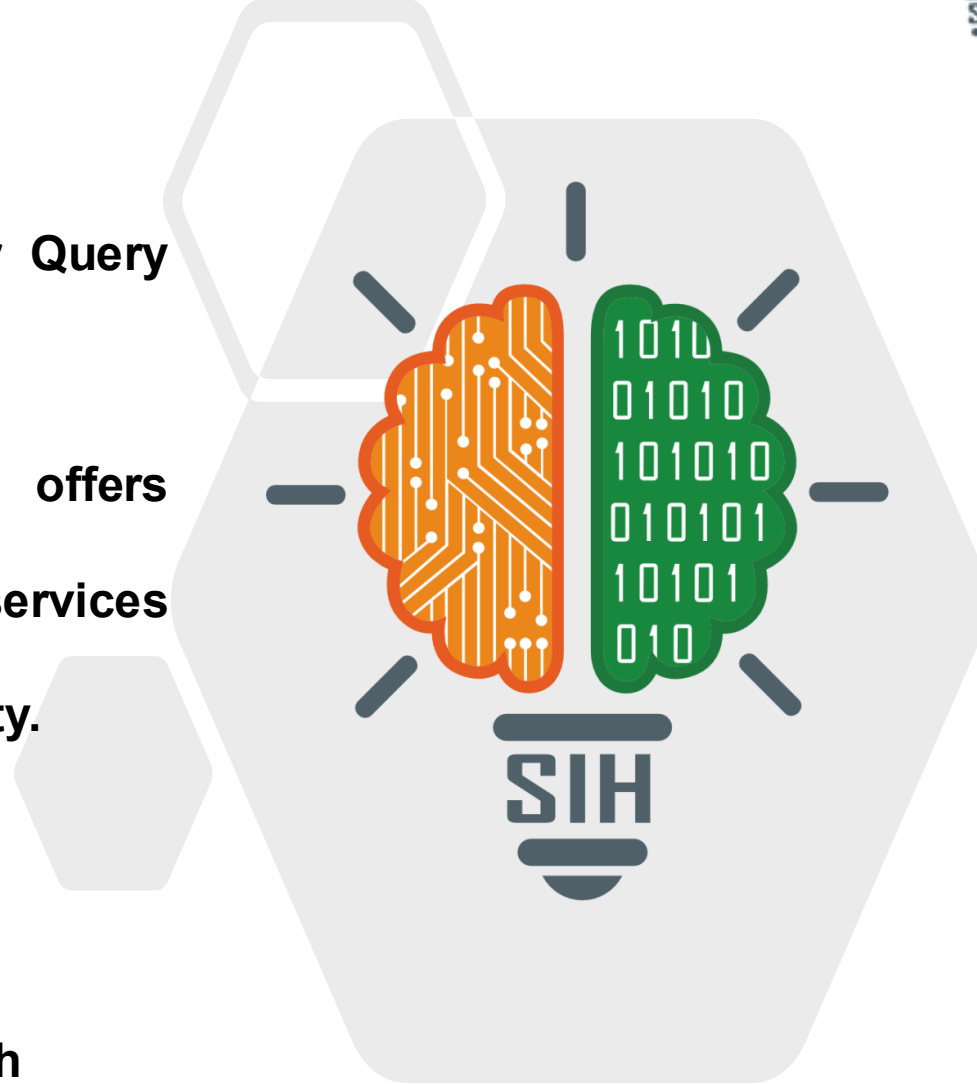


SMART INDIA HACKATHON 2025



- Problem Statement ID – SIH25076
- Problem Statement Title- AI based Farmer Query Support and Advisor System
- Theme- AI-driven farmer query support offers instant, reliable, and personalized advisory services to improve farming decisions and productivity.
- PS Category- Software
- Team ID-
- Team Name (Registered on portal)- Terra Tech



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- AI-driven **query & advisory system** for farmers (voice/text/image).
- Multilingual **chatbot with NLP** for local languages.
- Real-time data integration: **weather, markets, govt. schemes.**
- Personalized crop, irrigation, and pest management guidance.
- **Unique:** Voice-enabled, image recognition, offline SMS/IVR support.
- **Reduces dependency** on middlemen/extension officers.
- Ensures **faster, reliable, and localized solutions.**

- **Technologies:** Python, Java/Kotlin, React/Flutter, TensorFlow/PyTorch.
- **Databases:** Firebase/SQL + APIs (weather, govt portals, vision).
- Farmer input via **voice, text, or image**.
- NLP + AI model processes and retrieves advisory.
- Response delivered instantly in **local language**.
- **Follow-up option** for complex queries → expert escalation.
- System designed for **mobile-first + offline accessibility**.

- **Scalable** cloud-based solution, deployable across India.
- **Integrates** with govt. agriculture databases.
- **Challenges:** Low connectivity, adoption barriers, data accuracy.
- **Strategies:** Offline SMS/IVR, awareness & training programs.
- Supported by **Agri universities & Krishi Vigyan Kendras**.
- **High feasibility** due to growing smartphone penetration in rural India.
- Long-term **viability through govt. & NGO collaborations**.

IMPACT AND BENEFITS



- Provides farmers with **timely, expert-like advice** anytime.
- Ensures **inclusive access** for small and marginal farmers.
- Strengthens **decision-making**, reducing risks and crop failures.
- **Social impact:** Bridges knowledge gaps and empowers communities.
- **Economic impact:** Increases productivity and farmer profitability.
- **Market benefit:** Improves access to fair pricing opportunities.
- **Environmental benefit:** Promotes sustainable and eco-friendly farming practices.

- **ICAR** – Indian Council of Agricultural Research.
- **Vikaspedia Agriculture Portal** (Govt. initiatives).
- **FAO** – AI in Agriculture global reports.
- **IBM Watson Agriculture** case studies.
- Research papers on **AI & NLP in Agriculture** (IEEE, Springer).
- **Digital India & AgriTech policies** supporting AI adoption.
- **Global examples** of AI in precision agriculture.